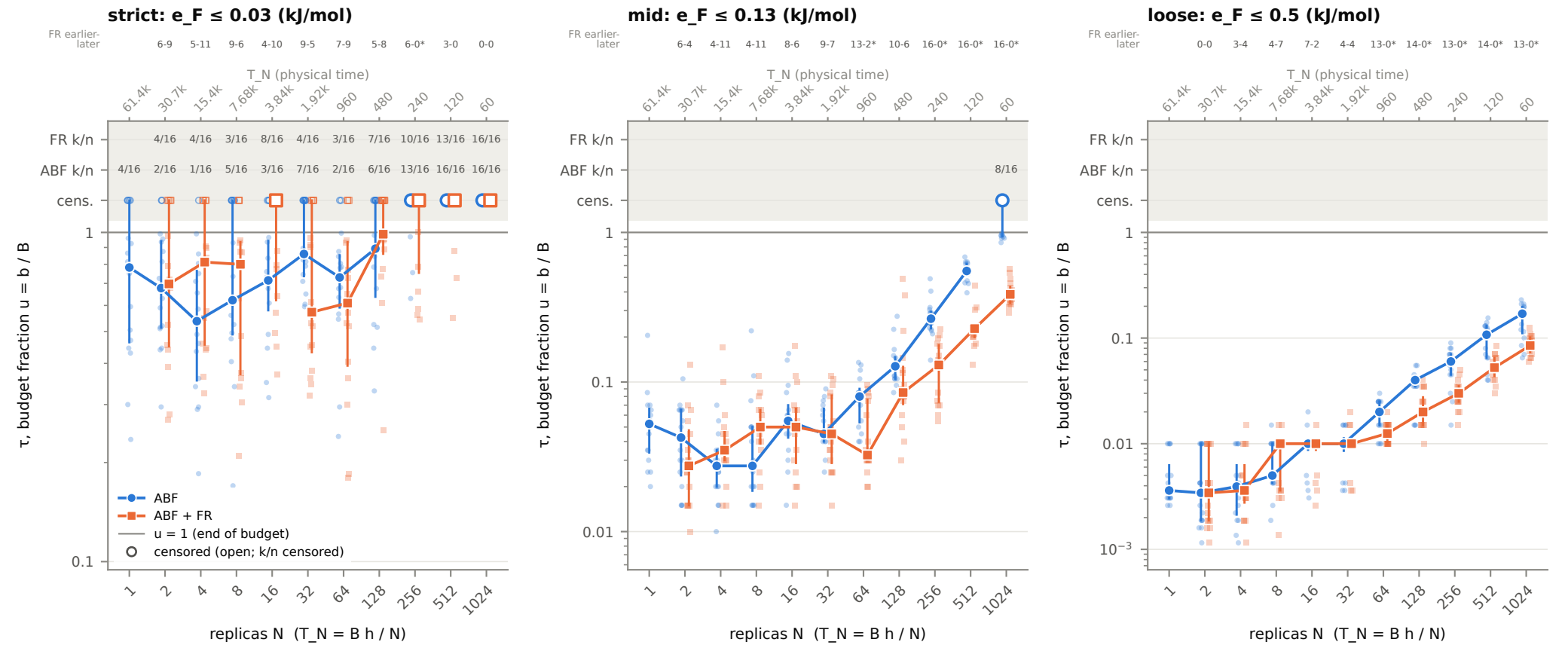


# ethane/LTA 150 K: persistent time to free-energy accuracy, budget fraction

Equal total budget per arm per seed:  $B = N \times n_{\text{steps}} = 3.072\text{e}+08$  walker-steps at every  $N$  ( $h = 0.0002$ ,  $T_N = B h / N$ ); 16 seeds per ( $N$ , method).  $\tau(\epsilon) =$  first save after which  $e_F \leq \epsilon$  at every later save; never met: censored ( $> T_N$ ,  $> 1$ ).



ethane/LTA 150 K · S4b\_tau\_free\_energy\_budget · status: all planned ( $N$ , seed, method) complete · FR is not defined at  $N = 1$  (ABF only, by design) · LTA  $N < 50$ : FR uses the finite- $N$  cap extension  $\max(1, \text{floor}(0.02 N))$ . Markers: seed median (large open marker in the 'cens.' band = median censored); bars: IQR; faint: seeds; censored seeds are open markers in the band, never success at  $T_N$ . Row above each panel: the PAIRED comparison at that  $N$ , seeds with FR earlier - FR later than ABF (finite beats censored, both censored = tie; \* = exact sign test  $p < 0.05$ ; medians, G and CIs in summary.json / tables.md). Budget axis: tau on the common budget grid (200 uniform + 24 log-spaced  $u$ , identical at every  $N$ ) so that  $N$  are compared at equal resolution. Source results/equal\_budget\_v2/lta\_T150/analysis/summary.json (generated 2026-10-10T07:14:18Z, eqb\_metrics/2, 10000 bootstrap resamples); plot\_synthesis/2.